

MUTUAL FUND ANALYTICS PLATFORM

BLUESTONE CAPSTONE PROJECT

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EXECUTIVE SUMMARY

The Mutual Fund Analytics Platform was developed as part of the Bluestock Capstone Project to provide a comprehensive analytics solution for mutual fund data. The project covers the complete data analytics lifecycle, including data ingestion, data cleaning, database creation, exploratory data analysis, performance analytics, advanced risk analytics, and dashboard development.

The platform processes multiple mutual fund datasets containing information related to NAV history, Assets Under Management (AUM), SIP inflows, investor transactions, scheme performance, portfolio holdings, and benchmark indices. The processed data is stored in a structured analytical environment and used to generate actionable insights.

Advanced analytics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, CVaR, Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, and Sector Concentration Analysis were implemented. Interactive Tableau dashboards were created to visualize industry trends and fund performance.

The project demonstrates how data analytics can be applied to financial datasets to support investment decision-making and performance evaluation.

PROJECT OBJECTIVES

The primary objectives of the project were:

- To ingest and consolidate mutual fund datasets from multiple sources.
- To clean and validate the data for analytical use.
- To build an SQLite database for efficient storage and retrieval.
- To perform exploratory data analysis and identify trends.
- To calculate key mutual fund performance metrics.
- To perform advanced risk analytics.
- To develop a fund recommendation system.
- To build interactive Tableau dashboards for visualization.
- To generate actionable insights for investors and analysts.

DATA SOURCES

The project utilized ten major datasets:

1. Fund Master Dataset
2. NAV History Dataset
3. AUM by Fund House Dataset
4. Monthly SIP Inflows Dataset
5. Category Inflows Dataset
6. Industry Folio Count Dataset
7. Scheme Performance Dataset
8. Investor Transactions Dataset
9. Portfolio Holdings Dataset
10. Benchmark Indices Dataset

These datasets collectively provide information regarding scheme details, historical returns, assets under management, investor behavior, and benchmark performance.

ETL DESIGN

The ETL (Extract, Transform, Load) process was designed to ensure data quality and consistency.

Extract:

Raw CSV files were collected from multiple mutual fund datasets.

Transform:

Data cleaning and preprocessing steps included:

- Missing value handling
- Data type conversion
- Date formatting
- Duplicate removal
- Outlier validation
- Category standardization

Load:

The processed data was loaded into an SQLite database using SQLAlchemy. Multiple dimension and fact tables were created to support analytical queries and dashboard reporting.

The ETL pipeline ensured that all datasets were consistent, accurate, and ready for analysis.



Data Cleaning & Validation

- Date parsing
- Missing value handling
- Forward fill in NAV data
- Standardization of transaction types
- Validation of AMFI codes
- Removal of invalid records

Database Design

The project used SQLite as the analytical database.

The database schema consisted of:

Dimension Tables:

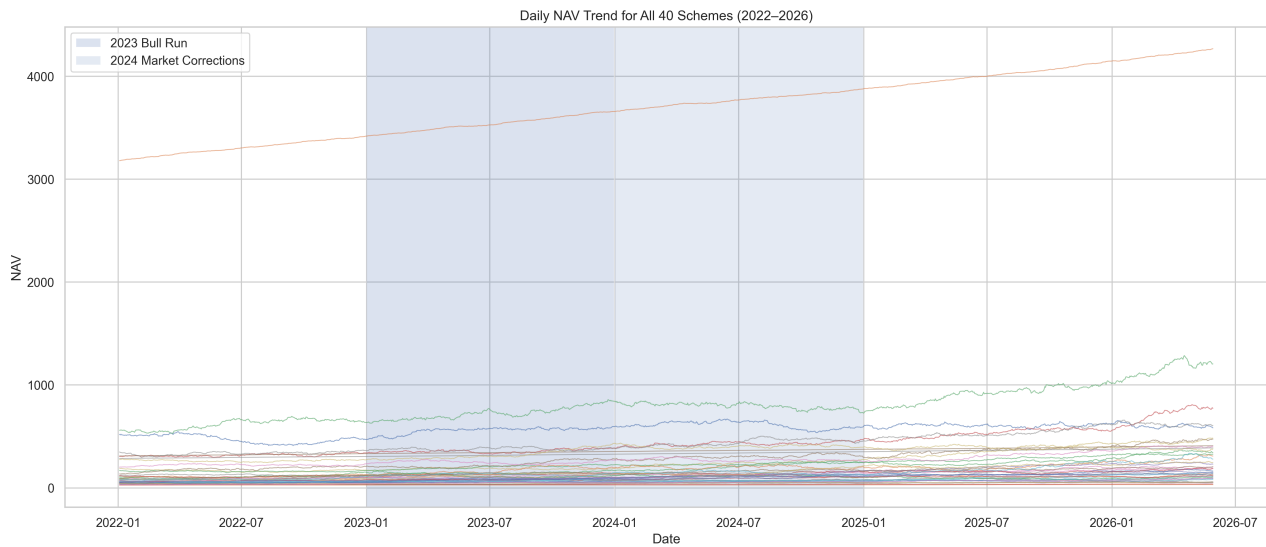
- dim_fund
- dim_date

Fact Tables:

- fact_nav
- fact_transactions
- fact_performance
- fact_aum

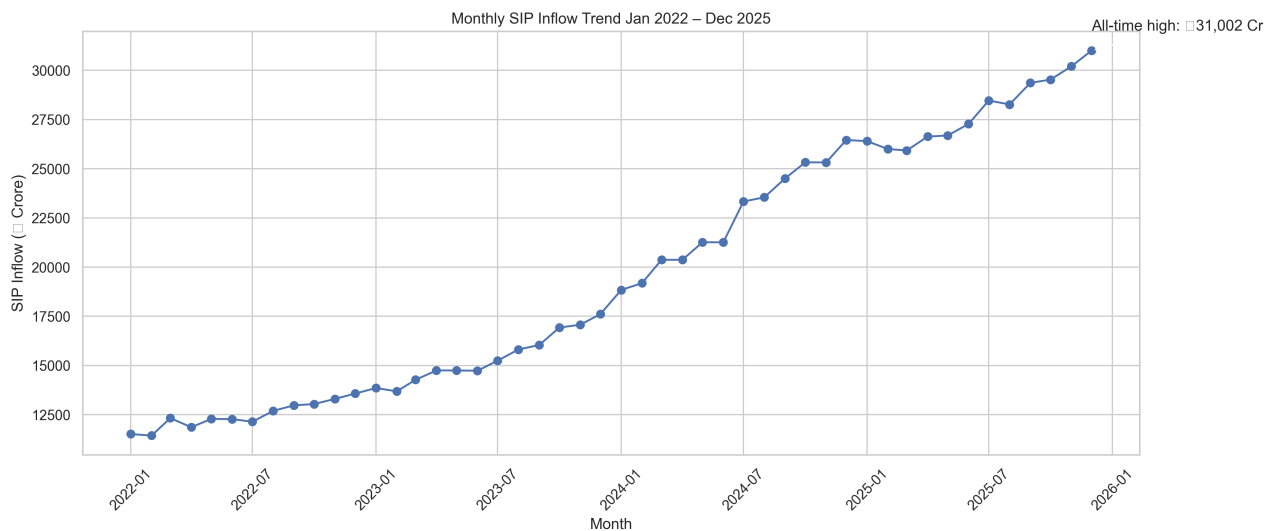
The star schema structure enabled efficient querying and simplified analytical workflows.

Exploratory Data Analysis (EDA)

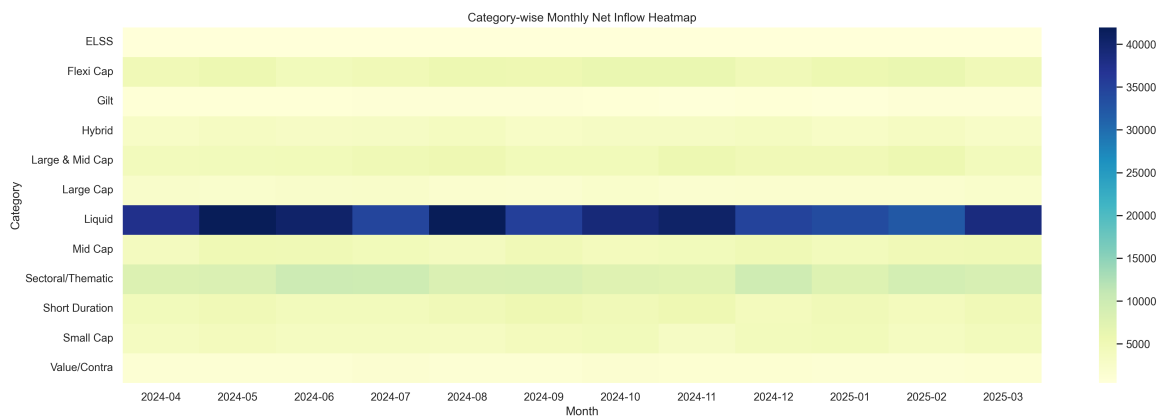


The NAV trend analysis showed consistent growth patterns across major mutual fund schemes. Long-term NAV movements indicated market expansion and increasing investor participation.

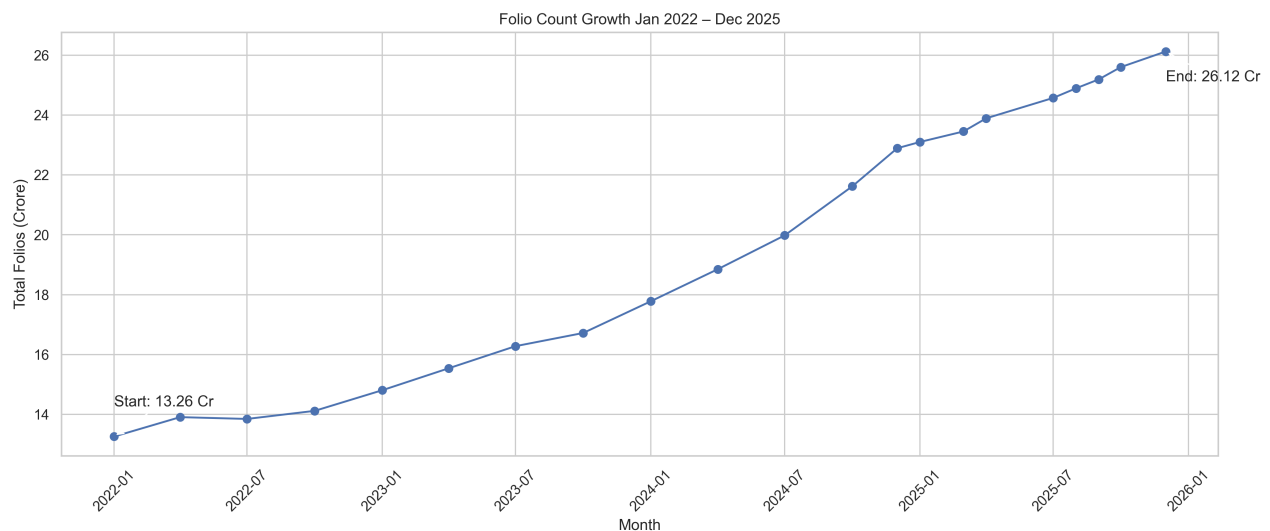
The analysis highlighted periods of strong growth and temporary corrections, reflecting changing market conditions.



Monthly SIP inflows demonstrated a strong upward trend. Increasing SIP participation indicates growing retail investor confidence and long-term investment behavior.

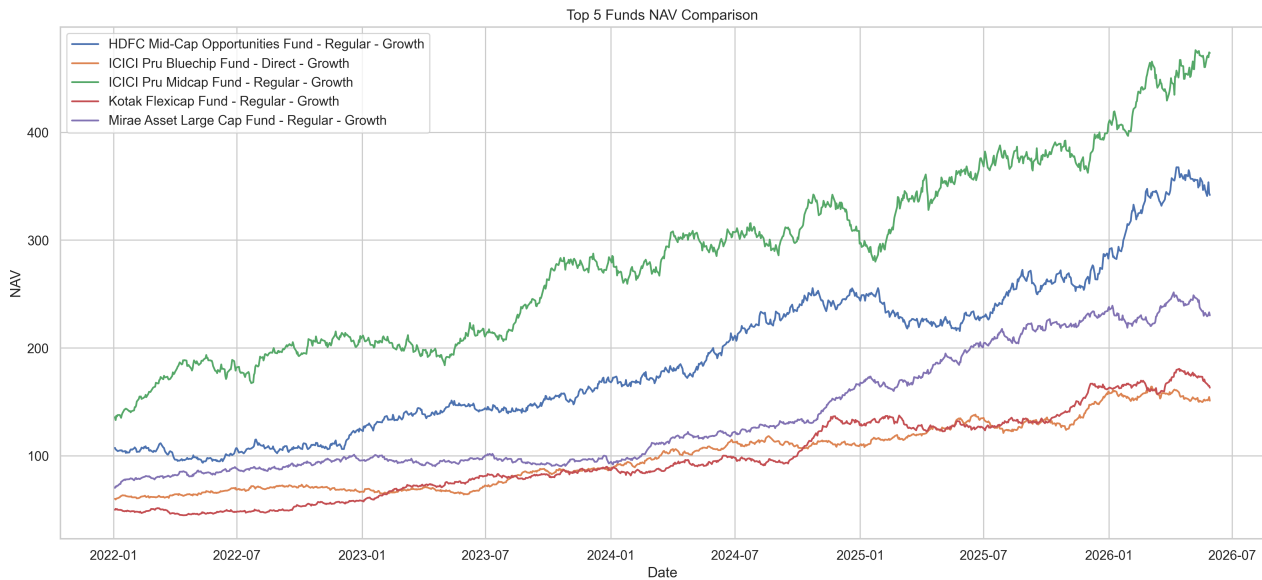


Category inflow analysis revealed that equity-oriented schemes attracted the highest investor interest. The heatmap clearly shows variations in fund inflows across different categories.



Industry folio counts increased significantly during the study period. This growth indicates broader market participation and increased awareness of mutual fund investments.

Performance Analytics

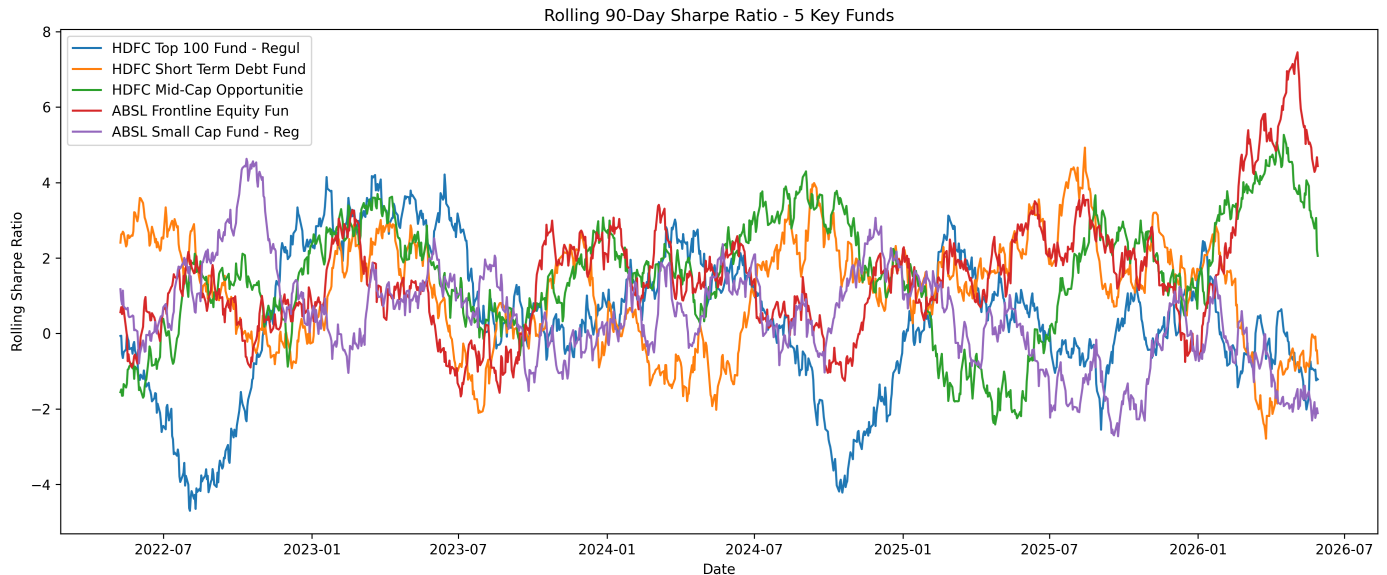


Several performance metrics were calculated:

- CAGR (1-Year, 3-Year, 5-Year)
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Benchmark comparison analysis enabled evaluation of fund performance relative to market indices. Funds with superior Sharpe Ratios demonstrated stronger risk-adjusted returns.

Advanced Analytics



Advanced analytics were performed to evaluate risk and investor behavior.

Implemented analyses included:

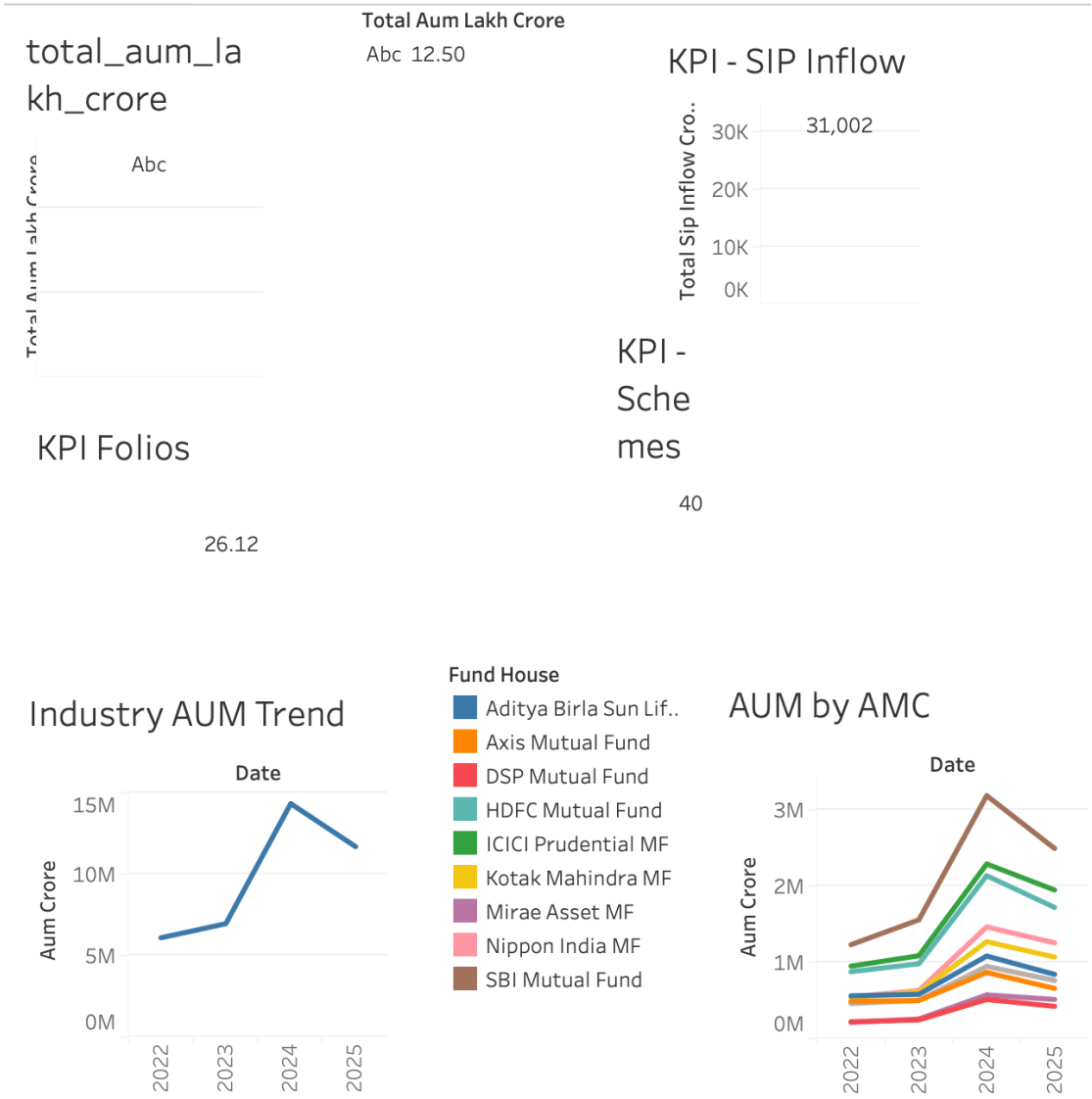
- Historical VaR (95%)
- Conditional VaR (CVaR)
- Rolling 90-Day Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Sector HHI Concentration Analysis

These techniques provide deeper insight into risk exposure and investment patterns.

Dashboard Overview

Industry Overview Dashboard showing:

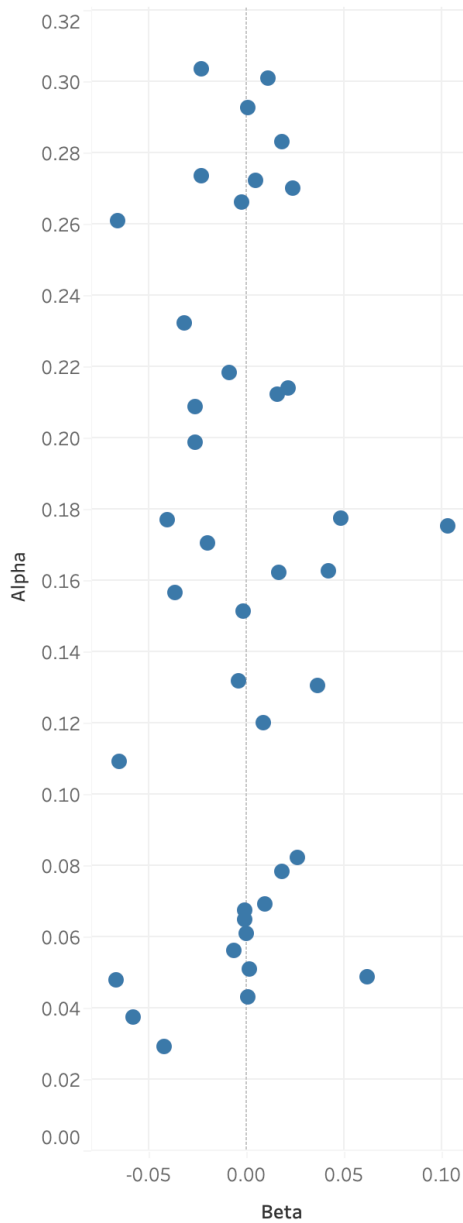
- AUM trends
- SIP growth
- Folio growth
- Fund house comparison



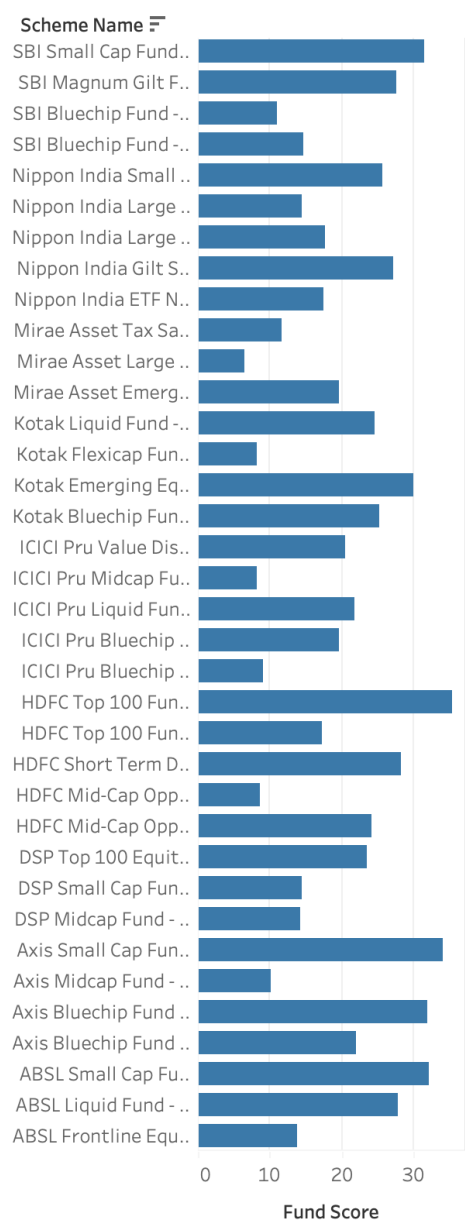
Fund Performance Dashboard showing:

- Top funds by score
- Alpha vs Beta comparison
- Performance insights

ALPHA VS BETA



TOP 10 FUNDS BY SCORE



KEY FINDINGS

- SBI Mutual Fund maintained the highest AUM among participating fund houses.
- SIP inflows showed continuous growth over the analysis period.
- High Sharpe Ratio funds consistently delivered stronger risk-adjusted returns.
- Most investors maintained regular SIP contributions.
- Certain equity funds displayed concentrated sector exposure based on HHI analysis.
- Rolling Sharpe Ratio analysis revealed changing risk-adjusted performance over time.

RECOMMENDATIONS

- Investors should consider risk-adjusted performance rather than absolute returns.
- SIP investing should be encouraged for long-term wealth creation.
- Diversification across categories can reduce portfolio risk.
- Sector concentration should be monitored regularly.
- Risk analytics such as VaR and CVaR should be incorporated into investment decision-making.

LIMITATIONS

- The analysis depends on the quality of available datasets.
- Market conditions may change over time.
- Historical performance does not guarantee future returns.
- Certain datasets were limited to available sample records.
- External macroeconomic factors were not incorporated into the analysis.

CONCLUSION

The Mutual Fund Analytics Platform successfully integrated data engineering, analytics, and visualization into a unified solution. The project demonstrated the use of modern analytical techniques for evaluating mutual fund performance, investor behavior, and risk characteristics.

The developed dashboards and analytics models provide meaningful insights that can support investors, analysts, and financial institutions in making informed decisions.

REFERENCES

Pandas Documentation

NumPy Documentation

Matplotlib Documentation

SQLite Documentation

SQLAlchemy Documentation

Tableau Public Documentation

AMFI India

NSE India